

Institute of Actuaries of India

Subject SA1– Health and Care

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:**i) Risk and rating factors****Individual IP**

Risk factors – health status, financial status, lifestyle, exposure to hazardous activities/substances, incentive to return to work, behavior wrt claiming [¼ per point, max 1½]

Rating factors – age, gender, smoking status, occupation, sum insured, location, replacement ratio, level of underwriting, length of deferred period [¼ per point, max 2]

Group MR

Risk factors – health status, financial status, lifestyle, exposure to hazardous activities/substances, ease of access to health care, behavior wrt claiming [¼ per point, max 1½]

Rating factors – age, gender, level of underwriting, industry, AML (Annual Maximum Limit) on claims payable, Network of providers, richness of the benefits, sub-limits/co-pay [¼ per point, max 2]
[Max 6]

ii) Risks in Pandemic

Increased claim outgo – both medical and economic challenges leading to higher claim payouts for both the products; this occurs in the short-term and likely in the medium to long term as the impact of a pandemic could be long lasting; impact will more pronounced from medium to long term perspective in the case of Individual IP with guaranteed rate structure [1½]

Liquidity risk - lower investment returns; depressed assets - depending upon the level of asset/liability matching there is likely increased risk of having to liquidate assets at lower value [1]

Solvency risk – depending upon how long/severe the pandemic lasts and the risk transfer arrangements that the insurer has in place, the solvency of the insurer could be threatened [1]

Counter-party risk – there is risk of third parties (eg. reinsurers, TPAs etc.) failing to honor their commitments [1]

Operational risk – difficulties to conduct business (eg. sales/underwriting/claim adjudicate) might expose the company to fraud/abuse risks [1]
[Max 4]

iii) Suitability of Reinsurance arrangements**a) Quota-share reinsurance**

- Structure with premium and claims ceded in the same proportion.
- Reinsurance cedes x% of all claims the insurer is still exposed to 1-x%.
- Ceding profit on ceded business as well – so, expensive
- Works better for reducing solvency capital requirement depending upon the regulatory requirements
- Simpler to administer

- Reinsurer would like to be hands-on and very involved
- Helps get technical support (underwriting, pricing, claims etc.) from the reinsurer if the insurer lacks own expertise/experience
- However, this option does not address the insurer's concern of catastrophic risk exposure [½ per point, max 3]

b) Individual Surplus reinsurance

- Another proportional structure
- Helps manage large claims volatility
- more suitable fixed benefit large face amount type products (eg. term insurance)
- Workable to some extent for IP – ceding policies above certain income benefit size
- Insurer is still exposed to higher frequency risk of a pandemic under the policies retained
- Doesn't work Group MR with indemnity benefit structure
- Overall, this option does not address the insurer's concern

[½ per point, max 3]

c) Per person excess of loss reinsurance

- Non-proportional arrangement ceding claims above certain size on a per person/policy per year basis.
- Cumulates multiple claims on the same person/policy and cedes the excess beyond an attachment point
- A lower attachment point will be akin to Quota-share
- so, reinsurer would not be keen on such arrangement without being hands-on and involved in the underwriting/pricing/claim decision making
- Catastrophic claims would likely be realized as a high frequency of claims, rather than a small number of high face amount claims.
- This option would not be quite appropriate to address the concerns.

[½ per point, max 3]

d) Per occurrence excess of loss reinsurance

- Non-proportional arrangement covering claims in excess of a threshold arising from a single event
- Suitability depends on the level of threshold and the treaty wording
- Again, lower threshold would be akin to Quota-share and the reinsurers would not be keen
- If, for example, the treaty defined a pandemic event as a single occurrence, then the option would be appropriate to address the insurer's concern
- If, for example, the occurrence to be one single claim/episode, then the option would not address the potential high frequency risk
- Likely come at a lower cost than alternative excess of loss options

[½ per point, max 3]

e) Aggregate excess of loss reinsurance

- Aggregate excess of loss covers all losses in excess of a threshold over a given timeframe
- There is likely an overall aggregate limit beyond which the losses fall back to the insurer
- So, the insurer is still exposed claims beyond the aggregate limit
- However, it is typically set at a level such that the chance of losses exceeding the limit is remote

- Likely more expensive than other XoL options
- Capacity/availability of such reinsurance is likely limited
- More appropriate for addressing the insurer's concern of catastrophic claims

[½ per point, max 3]

[Max 15]

iv) Alternate risk transfer arrangements

Catastrophe bond

- Insurer would sponsor an off balance sheet Special Purpose Vehicle (SPV) which would issue bonds to investors
- If a catastrophic event occurs, insurer would be able to use the invested capital to pay off claims, while investors receive coupons and the return of principal if no catastrophic event occurs
- This approach would be appropriate - It provides funding when catastrophic events occur and has low counterparty risk as funds are received initially. The insurer should consider the risks and regulatory scrutiny associated with SPVs

Participating in a pool

- Insurer would participate in a pool of catastrophic claims with other insurers and would share overall claims results
- This approach would not be very useful as every insurer in the pool would likely be affected by the catastrophic claims

Setting up own captive

- The insurer would set up a captive reinsurance company for risk financing purposes that can be based in a jurisdiction with favorable capital and/or tax treatment
- This would not be appropriate as Captives are meant for risk financing, not risk transfer. The insurer is still responsible for claims should catastrophe occurs.

[1 per point; max 5]

v) National catastrophe pool – Operating mechanism

- Contribution by insurers every year proportionate to the risk ceded to the pool
- Could be split by product/geographic segment
- Net earned premium could be a high-level proxy to the risk
- Alternatively, pandemic risk capital under risk-based solvency regulatory framework could be a proxy to determine the contribution
- More simply, a flat fee per insured member (and, split by product segment)
- Periodical review of the adequacy of the contribution rates
- Need to consider how the pool will be capitalized?
- Would it be open to external investors?
- Perhaps, through Securitization of the pool?
- Partial risk transfer to the Capital markets
- How any surplus will be handled?
- Reserving approach to pool liabilities

- If the claims exceed contributions and that the capital is wiped off, who bears the risk?
- Is there a bailout or guarantee from regulator that this capital loss will be funded?
- Government required to inject a one-off liquidity into the pool
- Or, will any surplus/deficit flow back to the participating insurers and/or any investors?
- How would this be administered?
- Potentially, leverage data from existing regulatory reporting
- The pool needs to define clear the policy wording and the types of claims admissible
- How do we define a catastrophe/pandemic?
- Any pandemic will likely affect all insurers, so, consider risk transfer arrangements
- ceding some risk to a supra national level agency
- Or perhaps, ceding the risk to a well-capitalized global reinsurer or reinsurance pool
- Is this pool optional or mandatory to all health insurers?
- Can an insurer decide to opt out and propose his own coverage?
- What is the process that the insurer must follow if it decides to opt out?
- Are the termination triggers fair to all participants?
- How are any surplus distributed? Or are these rolled over?
- If an insurer decides to opt out, should we attribute a fair sense of the surplus or not?
- Funds to be invested predominantly in fixed income – although if this pool is expected to be a long-term venture, it may consider a partial investment in equities to benefit from better returns.
- Need for liquidity is important and the pool may require paying cash calls
- Strong leadership and Governance oversight
- Strong risk management framework to mitigate risks and propose solutions
- Independent actuarial review/monitoring

[½ per point, max 10]

[40 Marks]

Solution 2:

i) Estimating future investment income

For all assets, make allowance for the expenses of investment. [½]

For fixed-interest securities, the income and redemption proceeds are known in terms of dates and amounts. [½]

An allowance should be made for any possibility of default (eg for non-government issues) and we normally ignore the possibility of earlier sale. [1]

There is no difference for index-linked stocks, apart from the need to build in an estimate of future changes in the index. [½]

For equities, the current dividend stream can be projected, taking into account the economic outlook and the progress of dividends generally. [1]

Assumptions on equity market values will be needed for anticipated sales in the future, if any. [½]

For property, rental income and capital growth can be projected, again taking into account the prospects for this asset class and the economy. [1]

Conservative assumptions should be used for property, due to the lack of marketability (if they are to be sold). [½]

Also, allowance could be made for particular aspects of properties, such as rental review terms, refurbishment costs and the risk of void. [½]

Money-market investments can be valued as an infinite stream of income based on short-term interest rates. [½]

Alternatively, if it is needed to pay imminent claims, it could simply be included as a lump sum income. [½]

A mix of these two methods might also be suitable. [½]

Overseas investments can be treated in the same way as domestic investments, with an allowance for exchange rates and the different risk of default. [1]

Allowance should be made for the volatility of income and capital values of all the above assets. [½]

The assumptions used for estimating the future asset income should be consistent with those used for estimating future liability cash flows. [1]
[Max 8]

ii)

f) Asset Valuation:

Assets in the Solvency II balance sheet have 3 component parts:

- investments (eg in equities, bonds, property, etc)
- reinsurance recoveries
- participations (ie companies in which the insurance company owns at least 20% of the voting rights). [2]

For non-linked assets, Solvency II replaces previous rules on admissibility limits (ie the extent to which certain assets could be taken into account in the balance sheet) and counterparty or asset exposure limits with a principles based test, known as the “prudent person principle”.

The “prudent person principle” requires, amongst other things:

- the assets to be appropriate for the nature and term of the liabilities
- assets to be chosen in the best interests of policyholders
- derivatives to be used only for reducing risk or efficient portfolio management
- assets to be “properly diversified”, eg by asset class, issuer, geographical area. [2]

Method of valuation

● Investments

Assets are required to be valued at market value, based on readily available market prices in orderly transactions that are sourced independently (ie quoted market prices in active markets).

If such prices are not available then mark-to-model techniques can be used –

provided these are consistent with the overall market-consistent (or “fair value” or economic value”) approach, ie the amount at which the assets could be exchanged between knowledgeable willing parties in an arm’s length transaction.

Unquoted equities, unquoted derivatives and property are examples of assets for which mark-to-model techniques may be required.

This was a significant change where book values (ie original cost, possibly with depreciation) were often still used under Solvency I. [2]

- Reinsurance recoveries

Recoveries expected from reinsurance are shown as an asset on the balance sheet, rather than as a reduction in gross liabilities.

This is a change from the previous regime where reinsurance was allowed for by showing liabilities net of reinsurance (ie by reducing the liabilities rather than by increasing the assets).

The fair value principle for valuing assets applies to these reinsurance recoveries.

Such recoveries must be adjusted to allow for the best estimate of expected losses due to the default of the reinsurer. This is calculated as the present value of the expected losses in each future year, and the adjustment is typically small as the probability of reinsurer default is normally considered to be low depending, on the credit rating and security of the reinsurer. [2]

- Participations

Participations will mainly arise when considering groups or where companies have outsourcing subsidiaries. Again, the overall market-consistent approach continues to apply. [1]

[Max 8]

Technical Provision

Technical provisions are based on a market-consistent approach ... [½]

... and should represent the amount that the insurance company would have to pay in

order to transfer its obligations immediately to another insurance company. [½]

Such liability values are rarely directly observable in the market, so they will need to be

modelled using market-consistent assumptions. [½]

The technical provisions consist of a best estimate liability and a risk margin. [1]

The best estimate liability is the present value of expected future cashflows (benefits

and expenses less premiums) based on best estimate demographic assumptions. [½]

Future premiums can be taken into account up to the “contract boundary”. [½]

For long term health insurance business this normally means the maturity or expiry date

of the contract. [½]

Allowance for future expenses needs to take into account both overheads and directly attributable expenses. No closure reserve is required. [½]

The assumptions should allow for all expected decrements and policyholder actions, including lapses, and should take into account expected future changes in health status. [½]

It must take into account all relevant available data, both internal and external, when arriving at assumptions that best reflect the characteristics of the underlying insurance portfolio. [½]

These calculations should ideally be performed on a policy-by-policy basis. However approximations are permitted and grouped model points can be used. [½]

For some liabilities, including financial guarantees and options, a market-consistent simulation or stochastic analysis may be the most appropriate calculation approach. [½]

Discounting takes place using the relevant risk-free interest rate term structure. [½]

The correct reference rate to use will depend on the currency and timing of the cashflow to be discounted. [½]

For long-term predictable liabilities, where the insurer can hold matching assets to maturity and so not be exposed to the risk of changing spreads, a “matching adjustment” can be applied to the risk-free discount rate. [½]

The matching adjustment applies to the impaired life annuity business and its addition, which must be approved by the regulator is subject to strict requirements in relation to the eligibility of the assets and liabilities. [½]

The matching adjustment is derived by taking the spread on the portfolio of matching assets and deducting the “fundamental spread”, an allowance for the credit risks retained by the insurer. [1]

This should address the concern of the finance director with respect to the P&L volatility. [1]

Alternatively a volatility adjustment can be added to the risk-free discount rate if the liabilities are not eligible for the use of the matching adjustment. [½]

The purpose of the volatility adjustment is to reduce the risk of forced sales of assets in the event of extreme bond spread movements. [½]

The use of the volatility adjustment is subject to prior supervisory approval and is subject to certain risk management requirements (such as a liquidity plan and sensitivity analysis). [½]

This should address the concern of the finance director with respect to the P&L volatility. [1]

Risk Margin:

The risk margin is intended to increase the technical provision to the amount that would have to be paid to another insurance company in order for them to take on the best estimate liability. [½]

The risk margin therefore represents the theoretical compensation for the risk of future experience being worse than the best estimate assumptions, and for the cost of holding regulatory capital against this. [½]

It is determined using the “cost of capital” method, ie based on the cost of holding capital to support those risks that cannot be hedged. These include all insurance risk, reinsurance credit risk, operational risk and “residual market risk”. [½]

The cost of capital rate charged for these projected capital amounts is suggested to be 6% pa. [½]

Alternatively, the cost of capital element of the technical provisions can be calculated using a simplified approach, eg by selecting a driver which has an approximately linear relationship with the required capital. [½]

[33 Marks]

Solution 3:**i) Reasons for the top-up product**

- Competitors may be offering higher AML plans in general
- Gap in the coverage for certain high cost procedures and the top up fills the gap here
- Response to general concern of increased health care costs triggered by an epidemic/pandemic
- Marketing strategy to differentiate from competition.
- Optional top-up might help a group's AML kept low (covered by the employer) and top-up option given to the members at their own cost
- so, flexibility/tailoring to need/affordability
- Regulator may mandate that some specific procedures be covered to a higher limit
- Existing members or those in the books for years may have old product/terms => there is a need to align with new market demand or terms.
- Part of the coverage may be excluded or covered under different insurance –
- for example, car insurance providing medical reimbursement in case of accident. The top up fills the gap as coverage there may be limited.

[½ per point, max 4]

ii) Pricing

- The base will be estimated on an estimate of (a) likely frequency and (b) likely severity.
- Have regard to the level of underwriting and the potential for anti-selection and allow for it
- If all members of a group opt for the product (compulsory), the chance of anti-selection low
- However, the groups that choose to go for the top-up product may have some level of sub-standard risks
- So, demand for top-up will likely be positively correlated to the risk – this will be more pronounced for individual than group
- will likely involve stochastic modelling and fitting of distribution of claims
- past data will not likely have claims beyond the current AML

- so, need to refer to industry experience of any such high AML products
 - such limits would be hit in the event of major medical/accident/critical illness
 - so, experience of such products could provide insight into the frequency
 - if this is happening in the backdrop of a pandemic, refer to data/research on potential severity
 - experience from a different market on similar lines of business (top ups) or a market with claims distribution spanning across higher AML could be used to benchmark
 - Reinsurer's expertise/rates will be benchmarked against internal view of the cost of the product
 - such experience needs to be tailored to local market having regard to cost of care differentials
 - The product design is another factor that may influence the pricing.
 - If there are reasonable waiting periods, policy exclusions or time bound restrictions in built, there may be reduced need for conservatism.
 - If the insurer will reinsure this product, the reinsurance pricing will need to be reflected.
 - Profit margins, allowances for expenses, cost of capital will need to be factored through.
- [½ per point; max 7]

iii) Types of reserves

- Outstanding Claims reserve – claims reported but pending as of the valuation date; may require adjustment based on historical experience in terms of how much of the pending claims get ultimately paid out (eg. pre-authorized claims may have an estimate of what could eventually be paid out) – for large claims there may be case level estimation required [1½]
 - IBNR – Incurred but not reported reserves – claims incurred prior to the valuation date but not reported– historical development pattern of claims to be considered using techniques such as chain-ladder to develop these claims into what would ultimately come through [1½]
 - Unexpired risk reserves – representing the risk for the unexpired period (period for which premium has been received in advance but not fully expired) – this is typically portion of the premium that relate to unexpired period pro-rate; may require adjustment taking into account seasonality of claims – that is, exposure to the risk claims is not uniform during the covered period, and, to the extent the unexpired premium is insufficient to cover the unexpired period of risk based on emerging experience – for example, the take-up rate is lower than expected and so claims will be higher than expected [2]
 - Catastrophe risk reserves – additional reserve to cover above normal claims during a catastrophe – it could simply be % of premium collected – more complex stochastic modelling calibrated to cover claims in say 1 in 50-year event [1]
- [Total, max 5]

iv) Controls to minimize anti-selection

- Introduction of waiting period – say 6 months before eligible to claim.
- Higher Co-pay at least for, say, the first year of cover – so, there is some level of skin in the game
- The top up product covers claim due to certain events/illnesses only
- For example, accidents and a set of critical illnesses
- Underwriting for individuals and groups below certain size
- Restrictions based on whether members have pre-existing conditions or not.
- Offer to groups only on mandatory enrollment of all members
- Require actively at work for groups at inception
- Exclude spouse and children covered in groups or individually underwrite them
- Pitch the AML for top-up not heavier than any other high AML products prevailing in the market

- Open to new members only or open to existing members if past claims can be looked at to understand profile of lives.
- Age restrictions or age – based rates to risk more appropriately.

[½ per point; max 5]

v) Cannibalization of existing products

- If the insurer is offering separate high AML products, then some of them might switch to low AML + top-up combination
- ... if that is less expensive or to reduce the cost of cover for the group asking the individual members to go for top-up at their own cost
- Similarly, some individuals carrying high AML products may switch to a low AML products and top-up only during those times that they think they will likely claim high – so, selective withdrawal from the existing product and selective entry into new product
- If the AML enhancer is compulsory at group level, there is little chance of cannibalization as the groups may not wish to shift out
- ...they may not see the need for such enhancer, nor will they be willing to pay extra premiums for these.
- if the product is offered only to new members, there is a risk of lapse or selective withdrawal so that members re-enroll to benefit from product limit.
- The cannibalization may be a beneficial thing if it occurs as it would mean that the old terms get replaced by better product and this may, in fact be to the benefit of the insurer.
- Some existing groups may switch to very restricted coverage (largely self-insuring) from the insurer and then top-up

[1 per point; max 6]

[27 Marks]
